# Research Fundamentals

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# Who am I?

2011 — **Bachelor's Degree (B.E.,)**
Information Technology Engineering
Nepal

2014 — **Masters Degree (M.Eng.,)**
Telecommunication Engineering
Thailand

2017 — **Doctoral Degree (D.Eng.,)**
Telecommunication and Artificial Intelligence
Thailand, Japan, USA

2022 — **PostDoc**
Artificial Intelligence
Thailand, Japan

2023 — **Chief, AI Research Center**
Advanced College of Engineering and Management
Kathmandu, Nepal

# 51-100
QS WUR Ranking By Subject

Inter-University Research Institute Corporation
Research Organization of Information and Systems
National Institute of Informatics

UNIVERSITY of WASHINGTON

# 76
QS World University Rankings

# 229
QS World University Rankings

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# Research Profile- Google Scholar

## Research Interests:

- Wireless Communication
- Artificial Intelligence
- Telecommunications
- Project Management



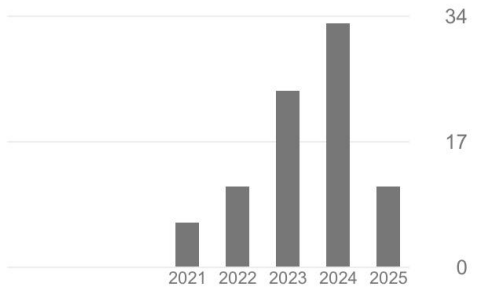
Bipun Man Pati

Advanced College of Engineering and Management
Verified email at acem.edu.np - Homepage

wireless communicaiton  spectrum sensing  Artificial Intelligence

Cited by

| | All | Since 2020 |
|---|---|---|
| Citations | 86 | 85 |
| h-index | 3 | 3 |
| i10-index | 2 | 2 |




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# Course Outline- Syllabus

Unit 1: Introduction (4 hrs)

Unit 2: Research Process Model (10 hrs)

Unit 3: Participants and Research Ethics (4 hrs)

Unit 4: Proposal Writing (4 hrs)

Unit 5: Report Writing (4hrs)

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# Course Outline: Introduction

1.1 What is research?

1.2 Research Aim and Objectives

1.3 Features of Research

1.4 Types of Research

1.5 The 6Ps of Research

1.6 Purpose of Research- reasons for doing research

1.7 Product of Research- outcomes of research

1.8 Research and Project

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# Course Outline: Research Process Model

2.1 Personal Experiences and Motivation

2.2 Literature Review

2.3 Research Question

2.4 Conceptual framework

2.5 Strategies

2.6 Data Generation Methods

2.7 Data Analysis

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# Course Outline: Participants and Research Ethics

3.1 Participants

3.2 The law and Research

3.3 Rights of People Directly Involved

3.4 Responsibilities of an Ethical Researcher

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# Course Outline: Proposal Writing

4.1 What is a research proposal?

4.2 Need of a Research Proposal

4.3 Components of a Research Proposal

4.4 A case study on any research paper/project

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# Course Outline: Report Writing

5.1 What is a research report?

5.2 Need of a Research Report

4.3 Components of a Research Report

4.4 A case study on any research paper/report

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# Course Outline- Books

1. Oates, B. J., Griffiths, M., & McLean, R. (2022). Researching information systems and computing. Sage.

2. Walia, A. M., & Uppal, M. (2020). Fundamentals of Research. Notion Press.

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# Course Outline- Evaluation

| Internal Evaluation | Weight | Marks | External Evaluation | Marks |
|---|---|---|---|---|
| **Theory** | | 30 | Semester End Examination | 50 |
| Attendance & Class Participation | 10% | | | |
| Assignments | 20% | | | |
| Presentation/Quizzes | 10% | | | |
| Internal Assessment | 60% | | | |
| **Practical** | | 20 | | |
| Proposal Report | 30% | | | |
| Project Presentation | 10% | | | |
| Final Project Report | 40% | | | |
| Completeness of Project | 20% | | | |
| Total Internal | | 50 | | |
| **Full Marks: 50 + 50 = 100** | | | | |

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# Practical Work

**Phase I:**

- The students are grouped in teams each containing at most 4 students.
- Each team chooses a problem to solve as their project/research work and they work in a team.
- They must define clearly what the problem is, justify why they choose the problem and how they will solve it and submit this as a proposal report (based on Unit 2 and 4).
- Each team presents their proposal orally.

**Phase II:**

- After the approval of their proposal, they start working on the project.
- Each team follows the research process studied in Unit 2 to do their project/research work.
- Students keep reporting their progress about the project/research work to their instructor.
- Students complete the project/research work, develop the final project/research report (based on Unit 2 and 5) and again present it orally.

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# Who are you?

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# Course Objective

1. What is research or project work,
2. How research is done,
3. Explain what research ethics must be followed during the research/project work,
4. How the research/project proposal is written, and
5. How to write final research/project report.

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